SDC HACKATHON 2026 · DOMAIN 1 · THE AIVERSECITY FUTURE-READY WORKFORCE DOMAIN

AURA-Workforce

A privacy-first career-transition planner: from “who I am today” to a sourced, sequenced plan for the AI era.

overinfinity · Ahmad Adam Hakimi bin Naeman · IIUM

Live prototype: aura-workforce.replit.app

THE PROBLEM

Workers are told to reskill, but not what to do on Monday.

56% of Malaysian professionals already use AI at work — only 26% have had formal training. (Hays Asia Salary Guide 2025)

MOSTI projects 600,000+ workers needing reskilling within 3–5 years.

Career advice is generic, static and unsourced: no way to see which roles are rising,

which skills transfer, how long a pathway takes — or where any claim came from.

The cost is paid twice: people buy courses that lead nowhere; employers retrain for roles that are closing.

WHO IT IS FOR

Four people, one moment of need.

The displaced worker — a procurement assistant whose routine work is being automated.

The switcher — a nurse or designer deciding where their existing skills can carry them.

The HR / L&D manager planning next year’s reskilling budget.

The TVET or university career officer advising students from a printed list.

The moment: the week a person learns their role is changing. They do not want a prediction; they want to know what to do next and whether to trust it.

WHAT IT DOES

Profile in. Ranked roles, gaps and a pathway out.

1 Intake — role, industry, experience. The model expands them into concrete, selectable skills.

2 Live trend layer — 300 recent public job postings, counted against each role's skills, stamped with source, fetch time and sample size.

3 Matching — deterministic, inspectable score: skills, transferable skills, sector, experience, demand.

4 Pathway — ordered steps, prerequisites respected, hours fitted to the learner's week.

5 SDG mapping — every role and step tagged to a sub-target.

6 Uncertainty — moderate (4-year) and aggressive (18-month) scenarios, not one forecast.

Two model calls. A closed world. No invented facts.

The model sees this person’s profile AND our catalogue — five roles, twenty learning steps.

It chooses, explains and sequences. Every role and module ID it returns is validated server-side;

unknown IDs are discarded. It cannot add a course, a provider or a statistic.

Its fit score is displayed next to the deterministic estimate — disagreement is visible, not hidden.

Beyond the catalogue, an exploratory tier proposes real occupations with a checkable reference

and an explicit “no verified pathway” badge.

Prompt injection is treated as data: delimited input, schema-validated output, vocabulary enforced.

THE SIGNAL LAYER

Updatable, dated, and honest about what it is.

Source: Arbeitnow public job board — 300 postings from the last 30 days, refreshed on demand.
Method: tokenised skill matching over full descriptions (naive substring matching inflated supply-chain demand from 21% to 45% — we found that by measuring, and fixed it).

Measured today: AI Operations 54% · Operations Research 35% · Supply Chain 21%

Every figure carries its source URL, fetch timestamp, window and sample size.
The interface states that this measures the direction of demand in a global board — not Malaysian vacancy counts.

SDG IMPACT FRAMEWORK

Baseline, KPI, and a target we can be held to.

Targets: 4.4.1 relevant skills · 4.3.1 lifelong learning access · 8.5.2 employment · 8.2.1 productivity.

Baseline: 56% of Malaysian professionals use AI; 26% trained. 600,000+ need reskilling.

Prototype today: 6 curated signals but only 1 updatable trend source; recommendations sourced ~60%.

KPI: % of role, skill and course recommendations carrying a verifiable source and date;

median time from profile start to a ranked pathway; number of independent trend sources.

Phase 2 target: 100% sourced and dated · median time-to-plan under 5 minutes ·

two independent trend sources, one Malaysian.

Hardest risk: the model inventing a job or a course.

Mitigation: closed catalogue, server-side ID validation, bounded fit, visible disagreement,
and an exploratory tier that cites an occupational reference or drops the suggestion.
Second risk: a global tech-oriented feed ≠ Malaysian vacancies — stated in the UI, Phase 2 fix.
Third: five roles cannot cover every occupation — the exploratory tier and 916-occupation search
make that limit visible instead of pretending otherwise.

Stack: React 19 · Vite · Tailwind · Express 5 · Zod · OpenAPI + Orval · DeepSeek · Replit · O*NET

Live: aura-workforce.replit.app Repo: github.com/adamnaemman/sdc-hackathon

WHAT COMES NEXT

Phase 2: breadth, Malayan context, and proof.

A Malaysian trend source (DOSM / local job postings) so the ranking reflects local demand.

Semantic skill matching with embeddings — lexical matching is the weakest part of the engine.

A counsellor view: cohort-level gaps, and plans that can be reviewed together.

Saved plans with explicit consent, one-click deletion, and no CV storage.

Self-hosted model on RunPod GPU, using the sponsor credit; the client is already OpenAI-compatible.

Team overinfinity · Ahmad Adam Hakimi bin Naeman · IIUM · SDC 2026 Domain 1